

Sequential Process Control

A process in which one event follows another until a job is completed

Characteristics

- 1.) Discrete loads
- 2.) Product output is in units (cans boxes)
- 3.) Different equipment modifies the product at each step
- 4.) Steps are staged (do step 1 before 2)

Applications

Assembly Line process

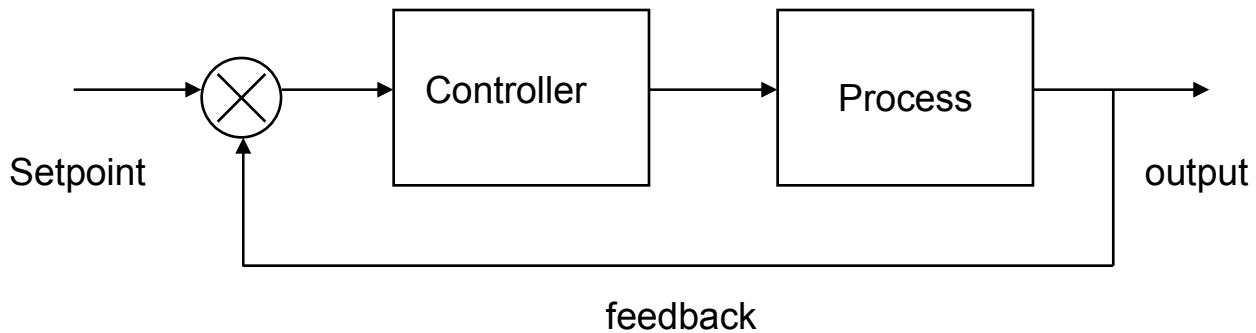
Conveyor systems

Robots

Power protection systems

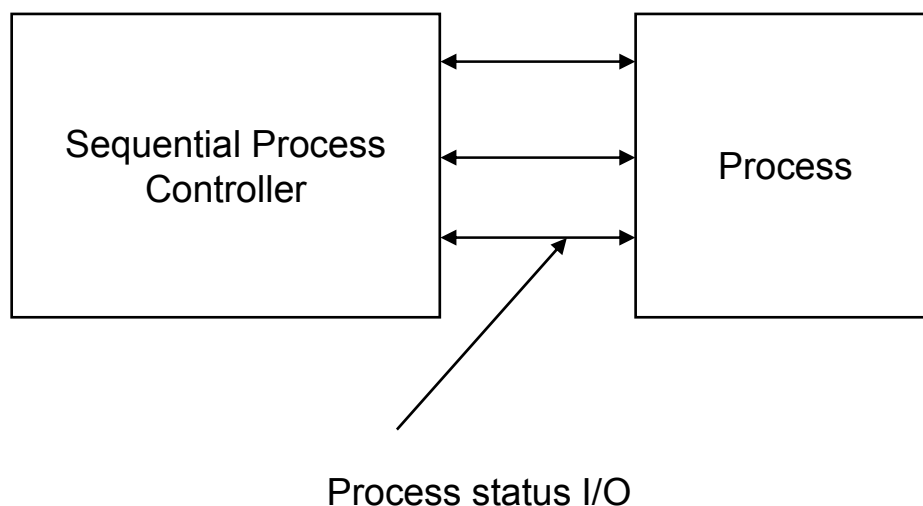
Motor starting and control

Sequential Vs Process Control



Feedback is continuous. Controller could be implemented with analog or digital control (DSP) methods

Status I/O typically bilevel (on-off) in nature



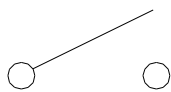
Devices Used in Sequential Control

Inputs: Manually Operated Switches

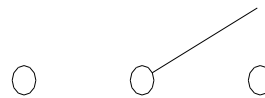
toggle switches- multiple poles multiple positions (on/off)

selector switches - sets different operating modes
(on/off/auto)

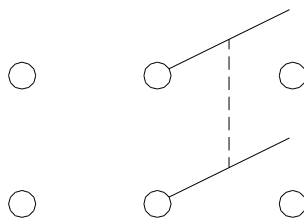
Schematic symbols



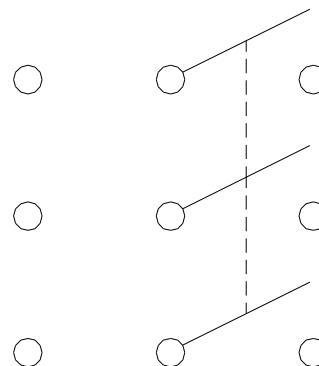
Single pole single throw
(SPST)



Single pole double throw
(SPDT)



DPDT
dashed line
indicates mechanical
links



3-pole double throw

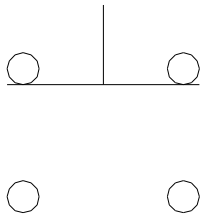
Drawing standard - switches shown in the unoperated or open position

Inputs-manually Operated Switches

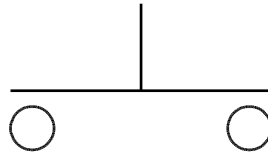
Push Buttons - momentary contact switches

Push buttons can be either normally open (NO) or normally (NC). Some types can be stacked to have multiple sets of NO and NC contacts

Schematic symbols



Normally closed
ganged with normally
open



Normally open
NO



Normally closed NC

Push buttons usually used
to start and stop pieces of
equipment in industrial
operation

Depressing the button causes all associated contacts to change state. Drawn in unactuated position.

Mechanically Operated Switches

Switches used to sense the operation of devices

Limit switches - switches that change contact state when there is movement. Detect if part of machine has reached a specific location. Usually physically linked to machine

Contact type - roller arm, wand type

Proximity type - detect ferrous and non-ferrous metals

Schematic symbols



NC limit
switch



NO limit
switch



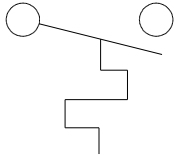
NO held
closed
limit switch

Limit switches can have multiple contacts of both the NO and NC type that change state when actuated

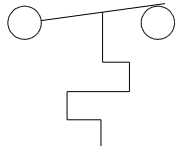
Drawing standard - switch drawn in the unoperated position

Mechanically Operated Switches

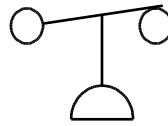
Used as sensors to detect conditions of operation



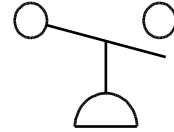
NO



NC

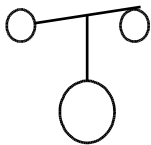


NC

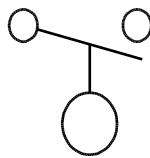


NO

Temperature switches
(similar to thermostat)



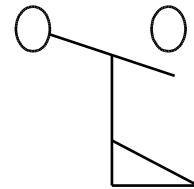
NC



NO

Level or float switches
Types exist that change
contact state for either rising
or falling level

Pressure switches
Types for both state
change on rising or
falling pressure

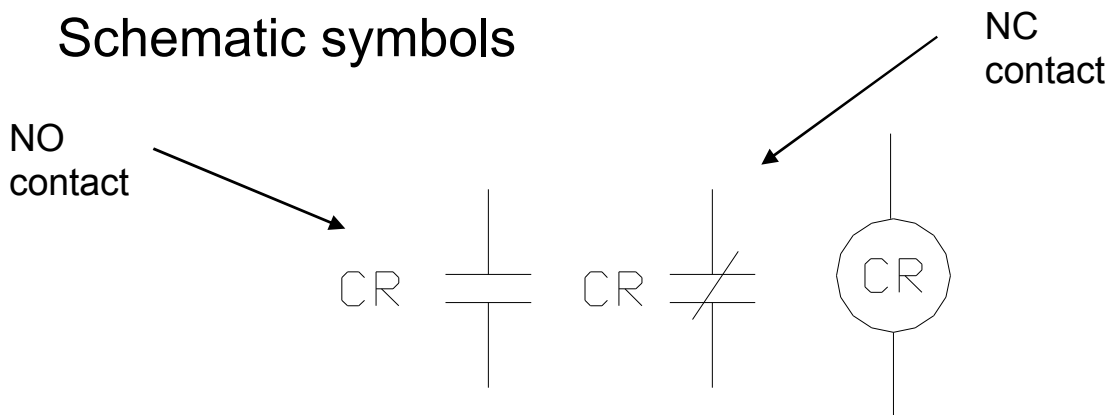


Flow Switch
(NO)
changes state
when level
reaches preset
(can produce
pulses proportional
to flow)

Control Devices

Control Relays - implement Boolean logic using electromechanical relays and contacts.

Schematic symbols



Drawing convention: control relay contacts shown with coil de-energized. Contacts associate with coil have similar identification

Coils and contacts need not be located together physically **on schematic**

Control Relay Characteristics

Coil voltage, current type (ac,dc), power consumption, pull-in power, I/V ratings of contacts, coil time constant

Control Relay Characteristics

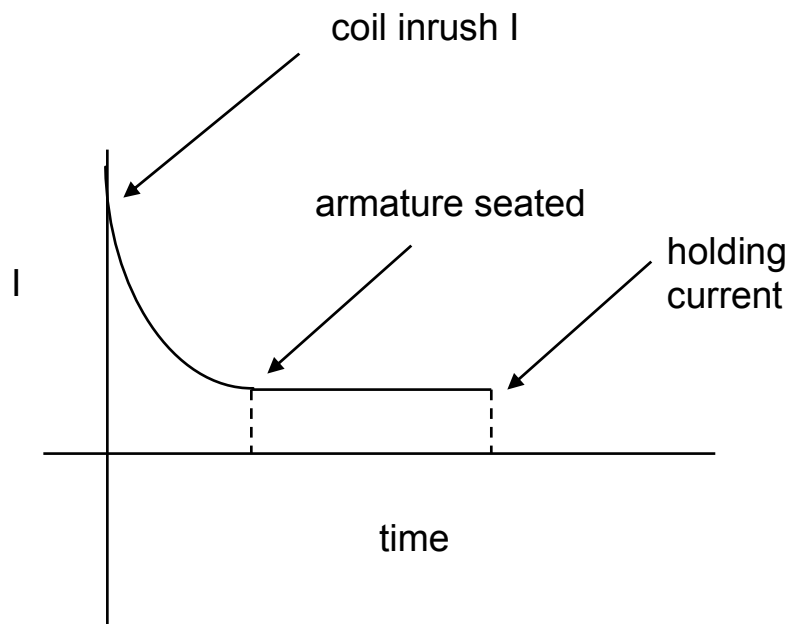
Coil voltage - operating range (+110% -80%)
typical values - 24, 48, 125 250 Vdc
48 120, 480 Vac

Low supply voltage causes relays to “drop out”

Coil Power - given in VA determine power capacity of supply.

typical 50 - 100 VA

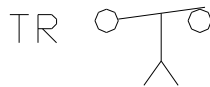
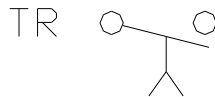
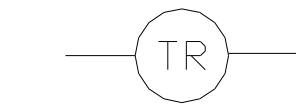
Pull-in Power - power necessary to move relay armature and close contacts. When requires 7-10 times rated (Reluctance high due to air gap)



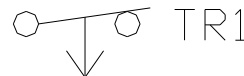
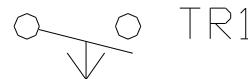
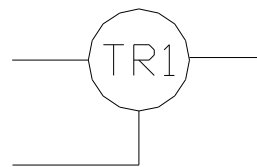
Electromechanical Timers/Counters

Timing relays used to delay contact state change by using electronically or electronically derived circuits

Schematic symbols: note either coil type is used



on-delay timer



off-delay timer

contact symbols determine the time of timer action

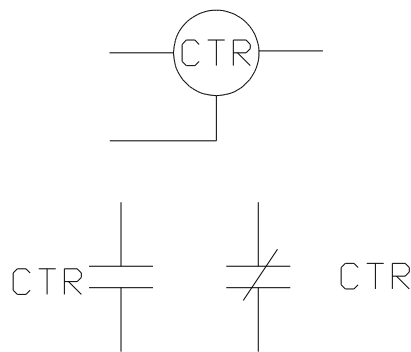
On-delay action - TR **energized**, contacts change state after set time interval

Off-delay action - TR **de-energized**, contacts change state after set time interval

Counters

Typically have three connections on schematic symbol for coil. Takes pulse input (contact closures) from other devices.

Schematic symbols



Contacts CTR change state after the preset number of counts are accumulated by CTR coil symbol

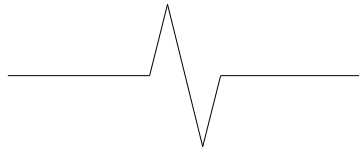
CTR device may also take a reset input that clears counter.

Final Control Devices

Solenoid- electromechanical device which uses a movable iron core to actuate another device

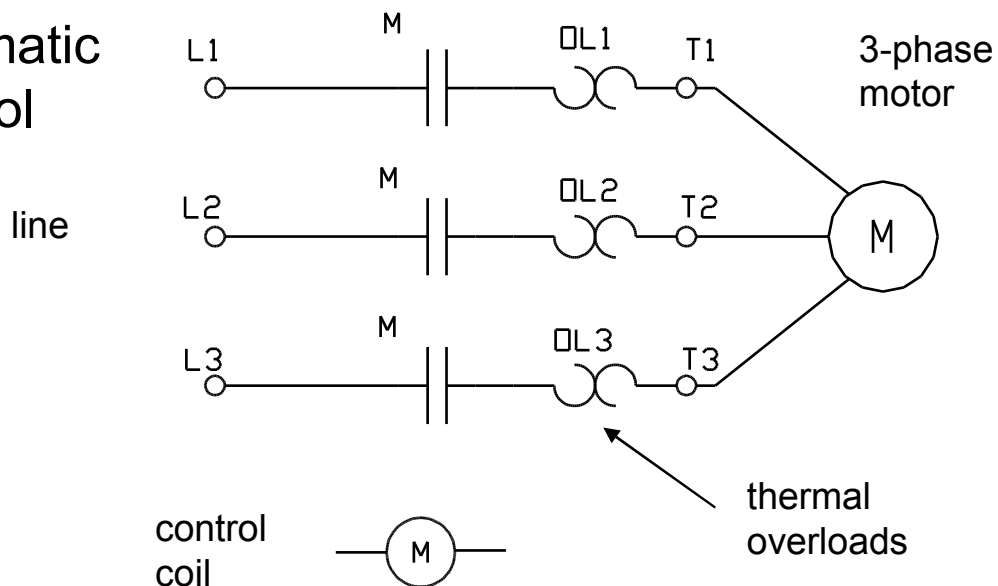
typical applications- valve control (liquid, pneumatic hydraulic)

Schematic Symbol



Motor Controllers- integrate switching with thermal overload protection

Schematic symbol



Basic Motor Control

Starter matched to motor by NEMA (National Electrical Manufacturers Association) size

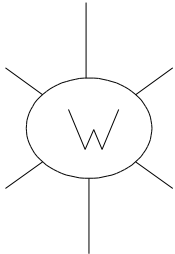
NEMA SIZE	MAXIMUM HORSEPOWER POLYPHASE MOTORS											
	FULL VOLTAGE STARTING			AUTO TRANSFORMER STARTING			PART WINDING STARTING			WYE-DELTA STARTING		
	200V	230V	460V 575V	200V	230V	460V 575V	200V	230V	460V 575V	200V	230V	460V 575V
00	1½	1½	2	—	—	—	—	—	—	—	—	—
0	3	3	5	—	—	—	—	—	—	—	—	—
1	7½	7½	10	7½	7½	10	10	10	15	10	10	15
2	10	15	25	10	15	25	20	25	40	20	25	40
3	25	30	50	25	30	50	40	50	75	40	50	75
4	40	50	100	40	50	100	75	75	150	60	75	150
5	75	100	200	75	100	200	150	150	350	150	150	300
6	150	200	400	150	200	400	—	300	600	300	350	700
7	—	300	600	—	300	600	—	450	900	500	500	1000
8	—	450	900	—	450	900	—	700	1400	750	800	1500
9	—	800	1600	—	800	1600	—	1300	2600	1500	1500	3000

Thermal overloads result from:

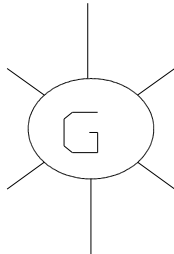
- 1.) Attempting to drive mechanical load greater than motor rating
- 2.) High inertia loads with long acceleration times
- 3.) Motor mechanically unable to turn
- 4.) Low motor terminal voltage
- 5.) Excessive starting and stopping (jogging)
(heat build-up due to high I)
- 6.) Loss of one of the three phases (single phasing)

Miscellaneous Devices

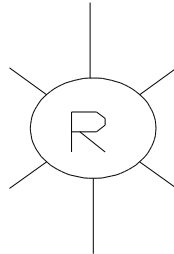
Panel Lights - Incandescent or LED



white



green



red



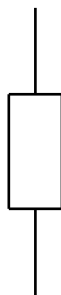
Size in amps indicated near symbol

fuse



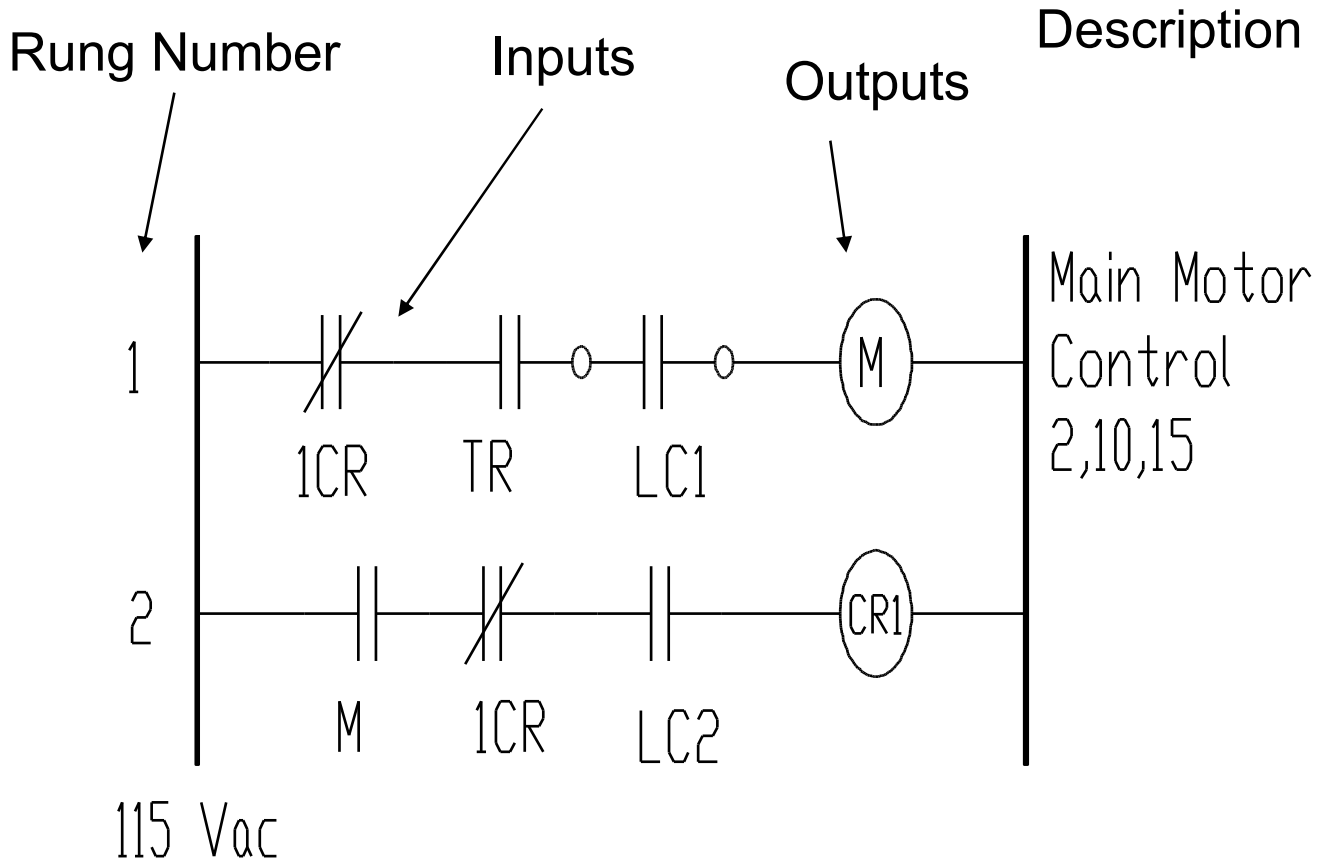
used to prevent moisture
buildup and component
icing in outdoor equipment

heater



Resistor

Ladder Diagrams



Circles on each side of contact indicate that it is physically separate from the other devices in the schematic

Number at each rung indicate the rung numbers where the output coil's contacts are located